proportional to the size of the object on which they act (Chapter 2), the waves will tidally distort *any* object by about 10^{-21} of its size. In this sense, 10^{-21} is the *strength of the waves* when they arrive at Earth.

Why are the waves so weak? Because the coalescing holes are so far away. The strength of a gravitational wave, like the strength of a light wave, dies out inversely with the distance traveled. When the waves are still close to the holes, their strength is roughly 1; that is, they squeeze and stretch an object by about as much as the object's size; humans would be killed by so strong a stretch and squeeze. However, when the waves have reached Earth, their strength is reduced to roughly (1/30 of the holes' circumference) / (the distance the waves have traveled).[1] For holes that weigh about 10 times as much as the Sun and are a billion light-years away, this wave strength is (1/30) × (180 kilometers for the horizon circumference)/(a billion lightyears for the distance to Earth) ≃ 10^{-21}. Therefore, the waves distort the Earth's oceans by 10^{-21} × (10^{7} meters for the Earth's size) = 10^{-14} meter, or 10 times the diameter of an atomic nucleus.

It is utterly hopeless to think of measuring such a tiny tide on the Earth's turbulent ocean. Not quite so hopeless, however, are the prospects for measuring the gravitational wave's tidal forces on a carefully designed laboratory instrument—a *gravitational-wave detector.*

Bars

Joseph Weber was the first person with sufficient insight to realize that it is *not* utterly hopeless to try to detect gravitational waves. A graduate of the D.S. Naval Academy in 1940 with a bachelor's degree in engineering, Weber served in World War II on the aircraft carrier *Lexington*, until it was sunk in the Battle of the Coral Sea, and then became commanding officer of Submarine Chaser No. 690; and he led Brigadier General Theodore Roosevelt, Jr., and 1900 Rangers onto the beach in the 1943 invasion of Italy. After the war he became head of the electronic countermeasures section of the Bureau of Ships for the D.S. Navy. His reputation for mastery of radio and radar technology was so great that in 1948 he was offered and accepted the position of full professor of electrical engineering at the